

Chenxin Zhong

Beijing

chenxin.zhong@outlook.com

Education

Southern University of Science and Technology

B.S. in Mathematics with Excellent Thesis

Sep 2016 – July 2021

Shenzhen, China

University of Regensburg

M.S. in Mathematics

Oct 2022 – Aug 2024

Regensburg, Germany

Research Experience

Bachelor Thesis: The Classification of Quadratic Forms over $\mathbb{Q}$ Dec 2020 – Apr 2021

Supervisor: Yong Hu

Shenzhen, China

- This thesis explores the Galois cohomology on the quadratic forms, and from the method, we can define some cohomology constants. Using the Hasse principle, we will show the quadratic forms over $\mathbb{Q}$ is classified by some constants by restricting to each $\mathbb{Q}_p$.

Master Thesis: Semistability Condition of Adelic Curves

Dec 2023 – Jun 2024

Supervisor: Klaus Künnemann

Regensburg, Germany

- This thesis explores the estimation of slope of the tensor product of Hermitian adelic vector bundles on adelic curves over K . By the method of GIT and Harder-Narasimhan filtration, we get a result such that the slope of tensor product of Hermitian vector bundles has an explicit lower bound for K of both characteristic 0 and p . Especially, when K is of characteristic p , we have that the tensor product of semistable Hermitian adelic vector bundles is semistable. Finally, we compare the tensorial semistability on vector bundles on algebraic curves X over K and on the corresponding adelic vector bundles over the corresponding adelic curves over $K(X)$ when K is of characteristic $p > 0$.

Professional Experience

Research Assistant

Jun 2021 – Jan 2022

Westlake University

Hangzhou, China

- I studied number theory and local class field theory under the supervisor Yigeng Zhao.

Research Assistant

Jun 2024 – Aug 2024

University of Regensburg

Regensburg, Germany

- Mainly literature review about the paper a modular analysis of FO transformation, study the modular analysis of FO transformation. And analysis different FO transformations of code based schemes they used.

Research Assistant

Apr 2025 – Now

School of Cryptography, University of Chinese Academy of Sciences

Beijing, China

- Mainly about lattice-based cryptography

Research Interests

Number theory: Non-archimedean Arakelov theory, integral and rational solutions of Diophantine equations, and height theory and stable conditions

Theoretical computer science: Algebraic and geometric complexity, post quantum cryptography and fully homomorphic encryption.

Awards & Honors

Excellent Student in Shude College	2nd Prize
<i>Southern University of Science and Technology</i>	2017
Excellent Student in Shude College	1st Prize
<i>Southern University of Science and Technology</i>	2018
China Undergraduate Mathematical Contest in Modeling	2nd National Prize
<i>Southern University of Science and Technology</i>	2017

Reading Seminars

Algebraic Number Theory, 2020
Algebraic Geometry, 2021
Complex Geometry, 2021
Galois Cohomology, 2022
 p -divisible Group, 2023
Adelic Line Bundles on Quasi-projective Variety, 2023
Height of K -semistable Fano Variety, 2024
The Arithmetic of Elliptic Curves, 2024

Other Interests

Language: English(C1), Chinese(Native), Germany(A1)
Programming: $\LaTeX$, Python, Matlab, Lean, C++